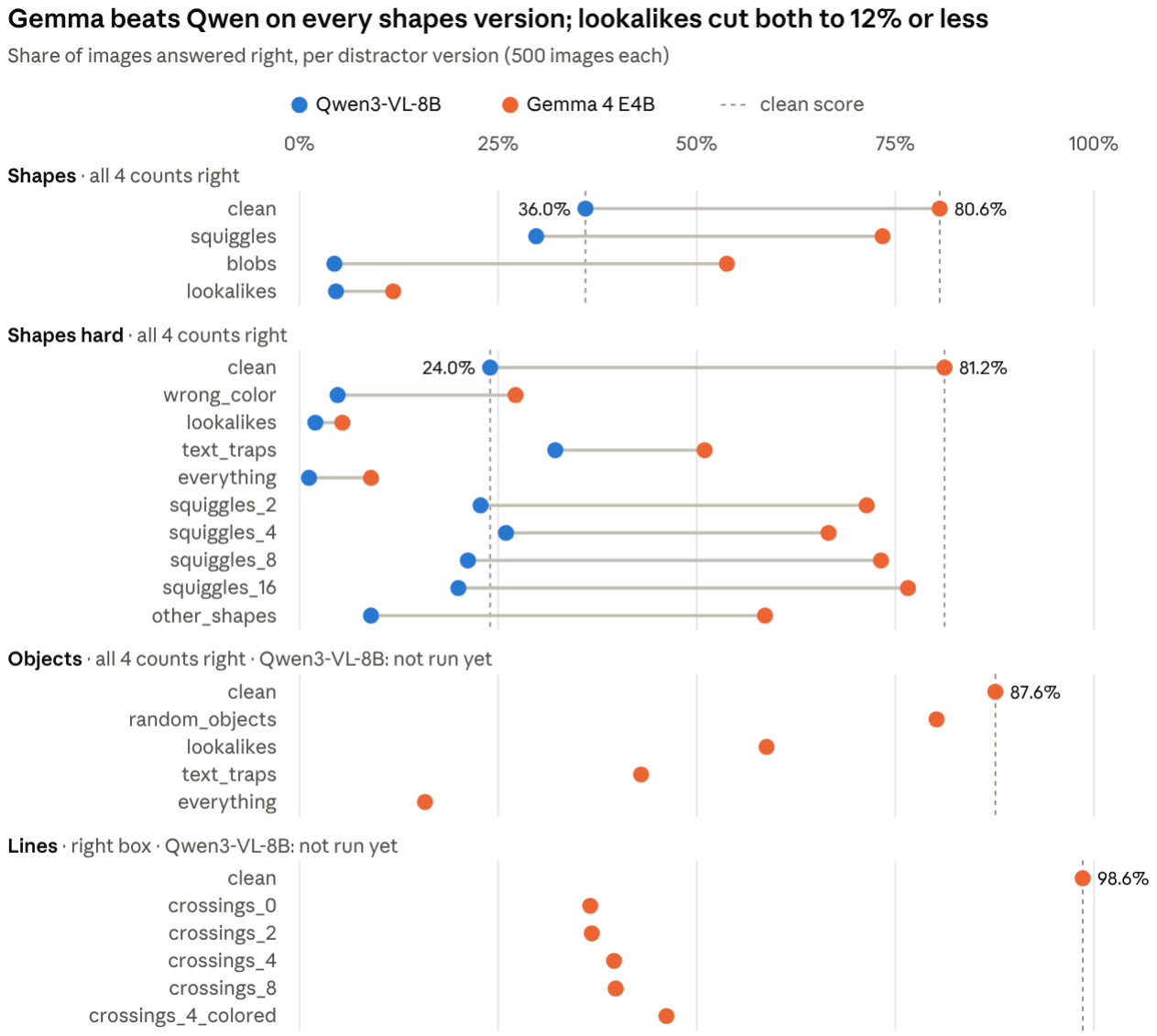
Study 3 results so far: Qwen3-VL-8B and Gemma 4 E4B

Nov 7, 2026 · @zylah

Where the runs stand

Two of the four Kaggle runs are done and clean: Qwen3-VL-8B part A (shapes and shapes-hard, 7,000 images, 3.7 h) and Gemma 4 E4B part 1 (all four datasets, 12,500 images, 7.5 h). Still to run: Qwen part B (objects, lines and the naming step) and Gemma part 2 (the naming step).

Accuracy by version



Kaggle run logs · Qwen3-VL-8B part A (version 10) and Gemma 4 E4B part 1 (version 8), 6–7 Nov 2026

Each row is one distractor version. The dashed line marks the model's clean score, so the distance from it is what that distractor costs.

What the numbers say

- **Gemma is the stronger counter.** On clean scenes it gets 80.6% (shapes) and 81.2% (hard) fully right, against Qwen's 36.0% and 24.0%. Qwen's misses are mostly triangles and squares (63.8% and 56.4% right on clean shapes).
- **Shape-like distractors hurt most.** Lookalikes cut Gemma from 80.6% to 11.8% on shapes and from 81.2% to 5.4% on hard. Gemma then counts too many shapes on 82–95% of those images.

- **Squiggles barely matter.** They cost Gemma 5–15 points on hard. For Qwen none of the four squiggle levels differs from clean (McNemar p ⩾ 0.06).
- **Text traps split the models.** Qwen does better with words in the image (32.2% vs 24.0% clean, p = 9e-5); Gemma drops 30 points to 51.0%.
- **Objects and lines (Gemma only so far).** Objects fall from 87.6% clean to 43.0% with text traps and 15.8% with everything. Lines fall from 98.6% to 37–46% once distractor lines are added. At crossings, 27–34% of answers follow the line that crosses the target.

Run quality and caveats

Both runs pass the checklist: no Traceback, every scorer exited 0, every image has a saved answer, and the speed lines are there (Qwen 1.9 s per image; Gemma 2.6 s, lines 0.7 s). Both used float16, greedy decoding and two T4 GPUs.

- **4 unparsed Qwen answers.** On 4 of the 5,000 hard images Qwen wrote only `circle: 4` and stopped, so three squiggle versions show 99.6–99.8% parsed. The scorer counts them wrong, and greedy decoding means a rerun gives the same answers.
- **Not measured yet.** Qwen on objects and lines, the naming step for both models, and the H1–H8 analysis, which needs all parts.